

Exploration of Acoustic Body Signals and Their Detectability Using 3D-Printed Ferroelectret Sensors

A Study on Signal Origins, Frequency Ranges, and Comparison with Existing Measurement Approaches

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1 Introduction

The human body emits acoustic signals that can provide valuable insights into a vast array of physiological and pathological processes. Ranging from the rhythmic beating of the heart to subtle sounds produced by the respiratory system and gastrointestinal tract or high frequency hits in joints, these signals offer a non-invasive window into the body's inner workings. Auscultation as one of the oldest diagnostic techniques, has been used since the days of Hippocrates (c.460 - c.370 BC) to assess the health of patients by listening to their internal sounds [1]. With the advent of modern sensor technology and advanced signal processing techniques the analysis of acoustic body signals has exceeded far beyond the capabilities of the traditional stethoscope and limited human hearing.

Differentiation of Acoustic body signals from force myography and mechanomyography This work explores the origins of various acoustic body signals and their respective spectrums. Providing excerpts of the physiological processes that generate these sounds and how they can be used for diagnostic analysis. In the second part of this work, an overview of in use and emerging sensor technologies for acoustic emissions is given, focusing on their working principles and frequency ranges. Lastly, the detectability of acoustically emitted signals using 3D-printed ferroelectret sensors is discussed and compared to existing measurement approaches.

2 Origins of Acoustic Body Signals

In this chapter the origins of acoustic body signals are explored. A focus on cardiac and vascular sounds is given first. Next gastrointestinal and respiratory sounds are discussed. Finally, acoustic emissions from joints, tendons and muscles are covered. For all signal types, the physiological processes that generate these sounds are described and characterised by their frequency content. If research exists on diagnostic capabilities or pathophysiological alterations of these signals it is covered in the respective sections.

2.1 Cardiac and Vascular Sounds

In a regular cardiac cycle the human heart emits sounds during two dominant events named the First Heart Sound (S1) and Second Heart Sound (S2) [2]. S1 is generated by the closure of the mitral and tricuspid valves at the onset of systole, while S2 stems from the halting of the aortic and pulmonary valves forcing a sudden deceleration of blood flow in the aorta and pulmonary artery, at the end of systole [2], [3]. Both actions form vibrations on the entire cardiac structure and surrounding tissues, which can then be detected as acoustic signals on the chest wall. Clinical practice has revealed a total of 5 positions on the chest wall where sounds can be best auscultated.

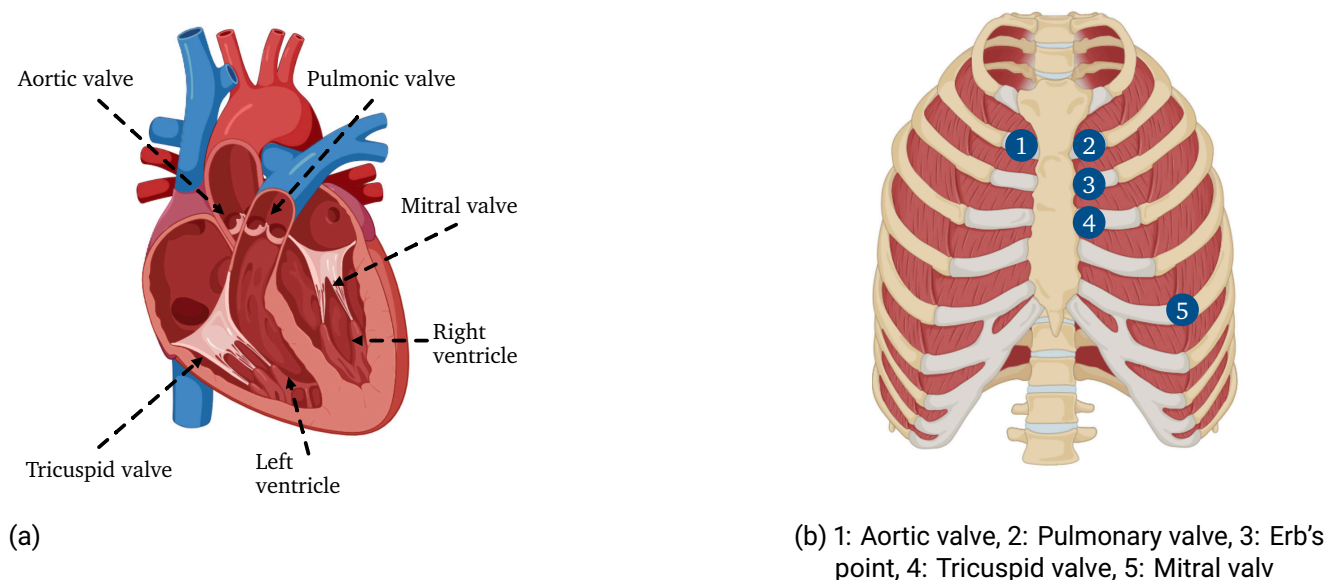
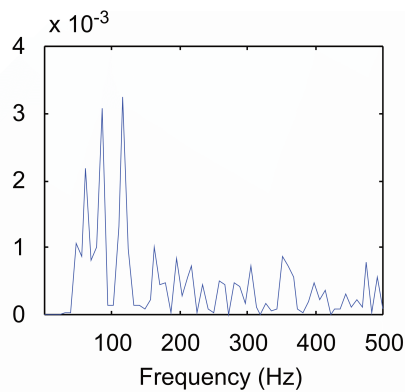
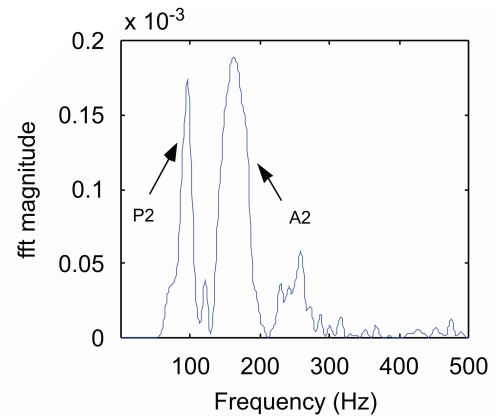


Figure 2.1: (a) Valves of the human heart and (b) Cardiac auscultation points on the chest wall.
Icons from biorender.com

Using Phonocardiography (PCG) the acoustic signals of the heart can be recorded and analyzed for diagnostic purposes. In the time-domain, S1 correspond to the QRS complex of the electrocardiogram (ECG) and S2 to the T-wave. A normal PCG signal has a frequency content in the range of 40 Hz to 200 Hz [3]. S1 typically lasts around 50 ms to 100 ms with a power spectrum in the range of 50 Hz to 150 Hz. S2 is shorter, lasting around 25 ms to 50 ms and has a higher frequency content in the range of 50 Hz to 200 Hz, which is expected as less blood is present in the cardiac chambers. The normal duration of systole is about 350 ms and that of diastole is 450 ms .



(a) Frequency spectrum for the sound S1



(b) Frequency spectrum for the sound S2

Figure 2.2: Graphs from S.M.Debbal et al. [2] , (b) shows the spectral decomposition of the sound S2, where the first peak corresponds to the aortic valve and the second to the pulmonary valve.

The spectral shape of S1 and S2 are influenced by body surface area, sex and blood pressure and ventricular mass. The frequency composition of S1 and S2 in combination with their temporal distance are valuable indicators for various cardiac diseases (Ventricular contractility and atrioventricular valve function, myocarditis, cardiomyopathy, myocardial infarction or heart failure disease). [3], [4]. Furthermore in pathological conditions additional heart sounds may be observed, such as the Third and Fourth Heart Sound (S3 and S4) or high frequency murmurs ranging from 600 Hz to 1000 Hz [5].

Diseases in the vascular system too produce acoustic signals. Turbulent blood flow in arteries caused by stenosis, aneurysms or arteriovenous malformations cause high pressure fluctuations in the post-stenotic region. Through the interaction of the vessel wall with the fluctuating blood flow, vibrations are generated and transmitted to the body surface [6]. The frequency content of these sounds is typically in the range of 100 Hz to 300 Hz, more severe stenosis (>70%) can produce sounds up to 1000 Hz [7]. Merely the existence of these sounds can be used for diagnostic purposes, but also their spectral content can be used to estimate the degree of stenosis [7].

2.2 Gastrointestinal Sounds

Phonoenterography (PEN) is the recording and analysis of acoustic signals produced by the gastrointestinal (GI) tract. Healthy PEN signals consist of a continuous isolated and short events in a frequency range of up to 1500 Hz with the majority of energy located between 50 Hz to 250 Hz [8].

Two approaches exist for interpreting PEN signals. The first is based on the analysis of the individual sounds and their categorisation [9]. The second approach is based on the analysis of the overall acoustic activity, like mean sound duration, silence between sounds duration, signal energy etc. The weighted combination of these features form the Acoustic Activity Index (AAI) [8]. Using the AII, the digestive activity of the GI tract can be assessed and thus used for diagnostic purposes and long term monitoring. [10] for example showed that AII can correlate physiological measures with motility or meal instances allowing diabetic patients real time consumption data.

2.3 Respiratory Sounds

The respiratory cycle consist of two main phases - inspiration and expiration. Normal adults breathe 16-20 Times per minute. A healthy lung produces sounds during both phases. In inspiration sound is generated through turbulent flow in the lobar and segmental airways, in expiration sound arises from more central airways. Frequency ranges from 50 Hz to 2500 Hz **rao_acoustic_2019**, [11], with the majority of energy in the range of 50 Hz to 500 Hz[12].

When deciding on sensor position for the detection of respiratory sounds, it is important to consider that the primary pathway for transmitted sound changes with frequency [12]. Lung parenchyma and tissue act as a low-pass filter and direct coupling to the chest wall is dominant. Thus transmission below 300Hz provides information mostly on lung parenchyma or thoracic cavity. Higher frequencies take the airway lumen route [13] and are thus best detected at the trachea or neck.

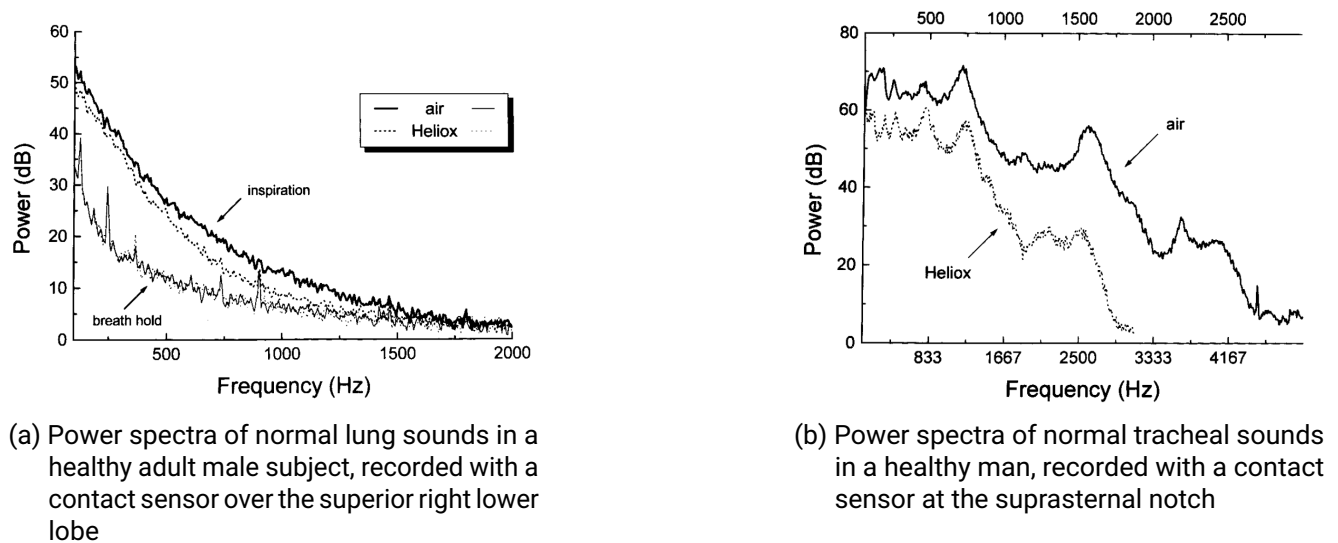


Figure 2.3: Graphs from H. Pasterkamp et al. [12]. Higher Frequency content can be observed in (b) due to the direct coupling to the airway lumen.

The additional graphs using Heliox show the parenchymal pathway is not affected by the change in gas density as sound is conveyed through the tissue.

Pathological conditions can produce additional adventitious sounds as listed in Table 2.1.

Table 2.1: Categories of Respiratory Sounds - adopted from [12]

Respiratory Sound	Mechanisms	Origin	Acoustics	Relevance
Basic sounds				
Normal lung sound	Turbulent flow vortices, unknown mechanisms	Central airways (expiration), lobar to segmental airways (inspiration)	Low-pass filtered noise (range < 100 to > 1000 Hz)	Regional ventilation, airway caliber
Normal tracheal sound	Turbulent flow, flow impinging on airway walls	Pharynx, larynx, trachea, large airways	Noise with resonances (range < 100 to > 3000 Hz)	Upper airway configuration
Adventitious sounds				
Wheeze	Airway wall flutter, vortex shedding	Central and lower airways	Sinusoid (range ~ 100 to > 1000 Hz; duration typically > 80 ms)	Airway obstruction, flow limitation
Rhonchus	Rupture of fluid films, airway wall vibrations	Larger airways	Series of rapidly damped sinusoids (typically < 300 Hz; duration > 100 ms)	Secretions, abnormal airway collapsibility
Crackle	Airway wall stress-relaxation	Central and lower airways	Rapidly damped wave deflection (duration typically < 20 ms)	Airway closure, secretions

* This table lists only the major categories of respiratory sounds and does not include other sound such as squawks, friction rubs, grunting, snoring, or cough. Current concepts on sound mechanisms and origin may be incomplete and unconfirmed

2.4 Joints and Tendons

Acoustic Emission (AE) of joints under load is widely considered to be a natural phenomenon [14]. As Emissions are generated by the movement of the joint surfaces, this dynamic approach is the main advantage over static mode examinations such as MRI or Ultrasound and thus can be correlated to joint angle and load. By means of placing a sensor on the skin over bony prominence and moving the joint, for example by carrying out a sit-stand-sit maneuver, acoustic signals can be detected. [15] AE Signals are generated by the interaction of the articular surfaces, ligaments and tendons. They consist of burst-type acoustic events with frequency components reaching up to 500 kHz [15].

In order to analyze the AE signal, it is common to describe hits. A hit is defined as a transient, burst like event satisfying a set of definition parameters. The most commonly used parameters are the amplitude, duration and frequency content of the hit [14]. A graphic overview is given in Fig. 2.4.

Another metric for assessing the health of cartilaginous joints is the analysis of the continuous activity summarized in the b-value [16].

$$\log_{10}(N_{AE}) = a - bA_{dB} \quad (2.1)$$

$$A_{dB} = 20 \log_{10} \left(\frac{V_{\text{peak}}}{V_{\text{ref}}} \right) \quad (2.2)$$

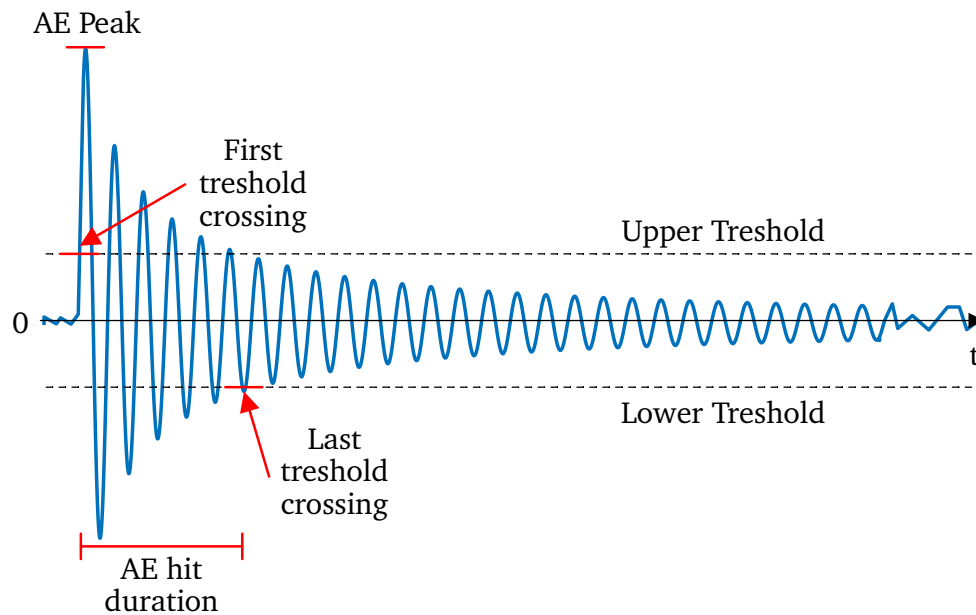


Figure 2.4: Graphic overview of the most commonly used parameters for describing and classifying an AE hit.

The b-value is derived from the field of seismology, expressing the logarithmic relationship between magnitude and frequency of a shockwave generated by an earthquake. In the context of joints, the coefficient b describes the decrease in the number of hits N_{AE} with an increasing amplitude A_{dB} serving as the detection threshold.

[16] showed, that the b-value is significantly lower in patients with an injured knee (1.46 ± 0.35) (torn anterior cruciate ligaments, torn lateral menisci, and sprained medial collateral ligaments) compared to healthy controls (1.92 ± 0.2), meaning that more high amplitude hits are present. [15] too observed a higher number of high amplitude hits in patients with osteoarthritis compared to healthy controls.

2.5 Muscle Sounds

2.6 Overview

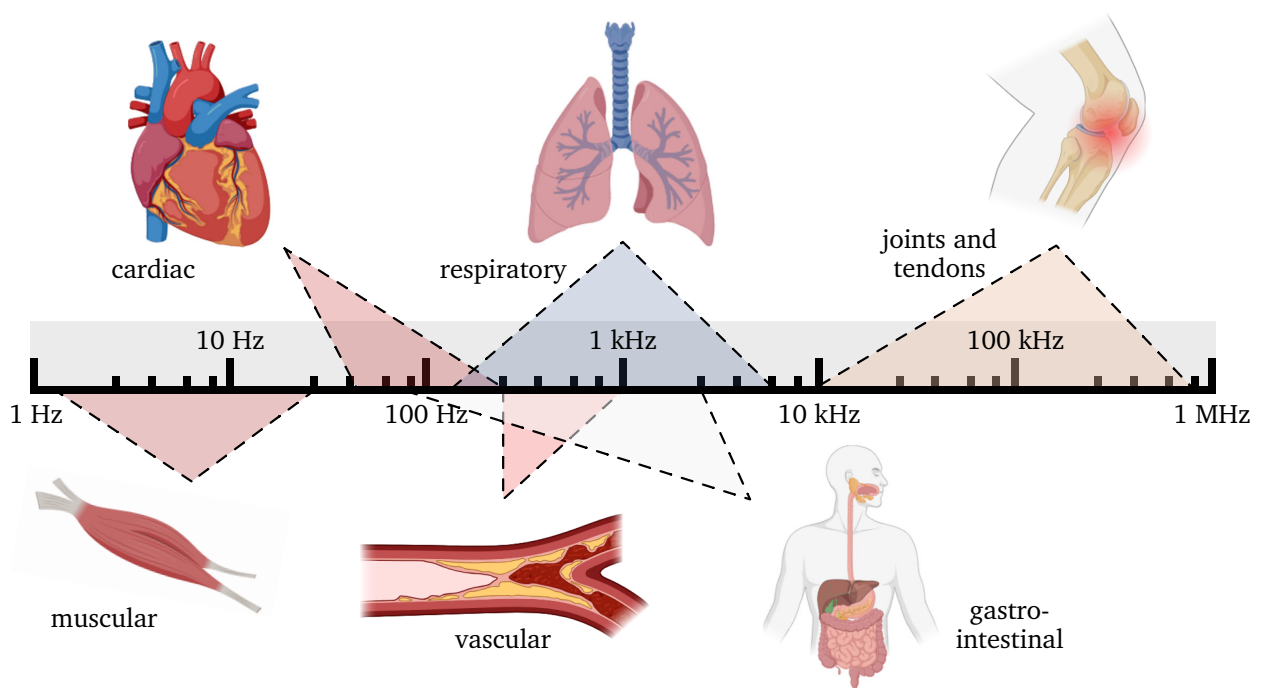


Figure 2.5: Overview of the frequency spectrum of acoustic signals.
Icons from biorender.com

3 Sensor Types

Due to the high acoustical impedance mismatch between fluid-like skin and air only a small portion of the acoustical energy is transmitted into the air, while the majority is reflected back into the body tissue. For this reason, many sensors are directly coupled to the skin, where they can detect the vibrations of the body surface with greater sensitivity.

3.1 Seismographic accelerometers Sensors

3.2 Piezoelectric Sensors

Piezoelectric (PE) sensors take advantage of the piezoelectric effect, which is the ability of certain materials to generate an electric charge in response to impact, vibrations and pressure. External forces on a PE material causes deformation and displacement of charge centers within the material resulting in polarization. By connecting electrodes on the surfaces of a piezoelectric material a sensor is built. A schematic of a PE sensor can be seen in Fig. 3.1 (d). Adjusting the thickness, shape or size allows for structural reconfiguration and a highly adjustable device that can be tailored to application specific requirements. Thus PE sensors can be designed to detect frequencies from few kHz up to several MHz [17]

3.3 Capacitive Sensors

Capacitive Transducers operate on the principle of capacitance change and are among the most widely used acoustic sensor. They are often optimized for the human auditive spectrum in the range of 20Hz to 20kHz. As can be seen in Fig 3.1 (a) the sensor consists of a fixed backplate and a flexible diaphragm, which form a capacitor. When an acoustic wave deflects the diaphragm, the distance between the two plates changes, leading to a change in capacitance. As the charge is constant, the voltage across the capacitor changes and can be measured [18] To provide constant charge on the electrode, a bias voltage is applied.

As an evolution of the capacitive microphone, Electret Condenser Microphones have a permanently polarized dielectric (electret) embedded in the diaphragm, eliminating the need for an external bias voltage. Electret-Condenser Microphones can be designed to detect frequencies from 10^{-3} Hz up to several 100 MHz [19] A schematic of the electret microphone can be seen in Fig. 3.1 (b).

3.4 Triboelectric Sensors

Triboelectric sensors are based on the triboelectric nanogenerator effect (TRENG) [20] enabling direct conversion of mechanical stimuli such as friction, vibration or deformation to electrical signal without external powering. Different topologies such as vertical contact-separation, single electrode, or lateral sliding mode exist [20]. The vertical contact-separation mode is the most commonly used topology and can be seen in Fig. 3.1(c). The sensor consists of two different materials with different triboelectric properties and dielectric constants ϵ_1 and ϵ_2 and thicknesses d_1 and d_2 . As soon as contact between the materials is established charge transfer occurs and a charge density σ_c is generated on the surface. When the materials are separated by a distance z , an electric field E_z is generated in the gap, which can be calculated using Gauss's law.

$$E_z = \frac{\sigma_1(z, t)}{\epsilon_1} \text{ and } E_z = \frac{\sigma_1(z, t)}{\epsilon_2} \quad (3.1)$$

The relative voltage difference between the two electrodes can then be calculated as follows:

$$V = \sigma_1(z, t) \left(\frac{d_1}{\epsilon_1} + \frac{d_2}{\epsilon_2} + \frac{z[\sigma_1(z, t) - \sigma_c]}{\epsilon_0} \right) \quad (3.2)$$

Under short-circuit condition, $V = 0$,

$$\sigma_1(z, t) = \frac{z\sigma_c}{d_1\epsilon_0/\epsilon_1 + d_2\epsilon_0/\epsilon_2 + z} \quad (3.3)$$

The displacement current density can then be obtained by applying Ampere's law on (3.3)

$$J_D = \frac{\partial D_z}{\partial t} = \frac{\partial \sigma_1(z, t)}{\partial t} \quad (3.4)$$

[21]

TENG sensors have been used in a wide range of medical sensing applications

3.5 3D-printed Ferroelectret Sensors

Ferroelectric Materials exhibit piezoelectric like properties such as that they convert mechanical signals (strain/displacement - stress/force) into electric signals (electrical discharge or Voltage). Unlike traditional piezoelectric materials which contain intrinsic dipolar units, electrets by definition are dielectric materials which can store electrical charge / polarization quasipermanently [22]. Typical materials are polymers such as polypropylene or fluoropolymers that contain microscopic air-filled voids. After fabrication a high voltage charging process deposits charges on the inner surface of voids which then behave like a dipole. The resulting sensor can be designed to have a high sensitivity and a wide frequency response, making them suitable for detecting a wide range of acoustic signals.

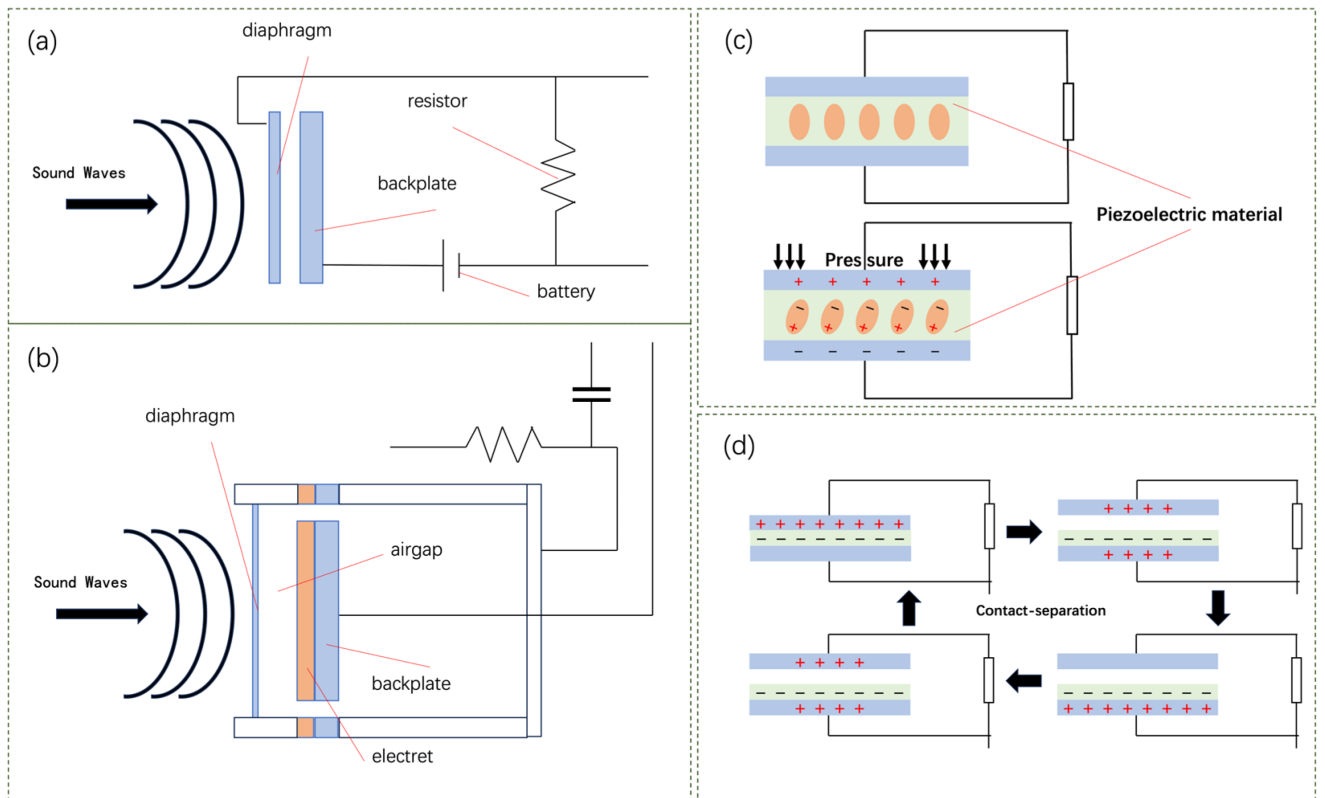


Figure 3.1: Types of acoustic sensors. (a) Schematic diagram of capacitive microphone. (b) Schematic diagram of the electret microphone. (c) Schematic diagram of piezoelectric sensor. (d) Schematic diagram of triboelectric sensor. Taken from [18]

4 Novel Measurement Approaches

Due to the high acoustical impedance mismatch between fluid-like skin and air only a small portion of the acoustical energy is transmitted into the air, while the majority is reflected back into the body tissue. For this reason, many new approaches use flexible substrates and sensors that are directly coupled to the skin, where they can detect the vibrations of the body surface with greater sensitivity. Another approach to harvest more acoustic energy is to use an array of sensors, enabling the capture of the spatial distribution and . A third attempt at producing better signals, external stimuli are introduced into the body.

4.1 Array sensed approaches

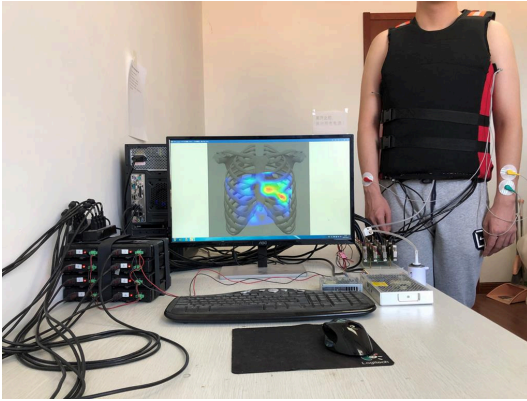
Array sensing is a technique that uses multiple sensors arranged in specific patterns along the body to capture acoustic signals.

4.1.1 Cardiac and Vascular Sounds

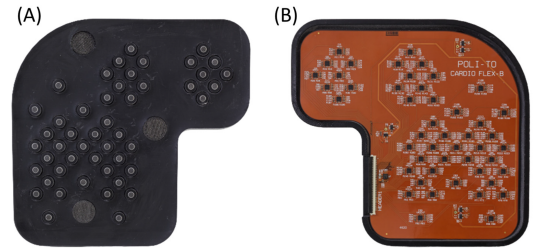
[23] developed a 72-channel MEMS based wearable vest for detection of cardiac sounds. By mapping the acquired signals to the chest wall, the spatial distribution of S1 and S2 sounds could be acquired and the best auscultation spot individually extracted. Additionally the trend of heart sound modulus on the chest surface could be observed over time. Guo et al. proposes, that the "technique may enable physicians to discover abnormal vibration occurring in the heart, such as valve disease, hypertension, and abnormal vascular reconstruction".

In another study N. Giordano et al. implemented a flex-PCB based 48-channel condenser microphone array for the detection of cardiac sounds and progression of chronic heart failure (CHF). In a n=42 volunteer study they showed that using their device all subjects were able to record signals of good to excellent quality by themselves. [24] believe that in the future, the device could be used for home monitoring of CHF patients.

[25] transformed the signals of a 2x2 MEMS microphone array placed on the chest wall through beamforming into a single signal to strengthen SNR. Using the newly acquired sensing capabilities M. Akay et al. showed that they can discriminate between patients with coronary artery disease (CAD) and healthy controls as well as patients treated with stents. Their findings indicate, that their approach can be used to continuously monitor stent placement noninvasively.



(a) 72-Channel MEMS based wearable vest. A visualization of the regional sound modulus is shown on the screen [23].



(b) PCB-Based Sensor array using 48 condenser-electret Microphones[24].

Figure 4.1: Multi Sensor AE devices for monitoring cardiac sounds.

4.1.2 Respiratory Sounds

In 2019 [26] proposed a Resonant-Microphone-Array-Based Wearable Stehoscope. Using 7 custom-designed MEMS piezeletric microphones A. Shkel et. al makes use of specifically tuned resonators ranging from 175 Hz to 700 Hz to allow for hardware-level spectral filtering. With improved SNR in each frequency band it allowed for lower noise floor and improved noise rejection.

Inspite of the large interference by cardiac sounds, wheezing could be effectively monitored at bronchial level instead of the standard tracheal measurement. This is possible as cardiac sounds have a distinct distribution of spectral peaks (As described in 2.3), that can be filtered by using the proposed resonant microphone array.

4.2 External stimuli

External introduced signals have the advantage of providing a fixed input signal instead of the available intrinsic sounds which are non-stationary and not repeatable.

4.3 MEMS approaches



5 Discussion



6 Conclusion

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